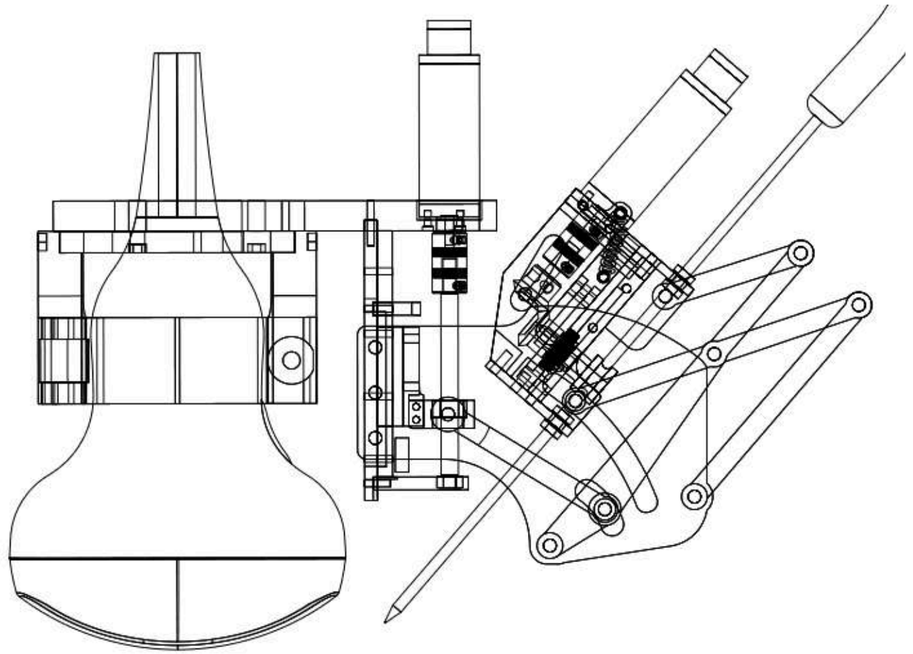
PROJECT 3

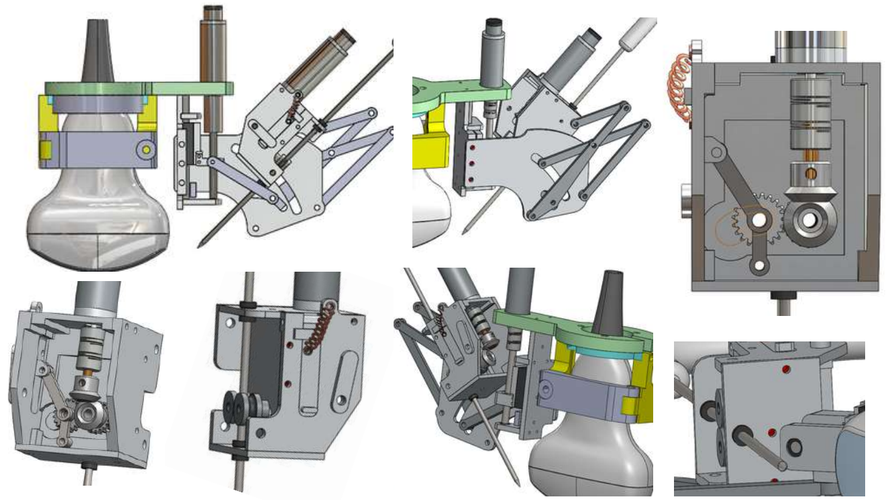
NEEDLE INSERTION MECHANISM



RCM BASED NEEDLE INSERTION MECHANISM

BRIEF BACKGROUND

This project introduces an improved Remote Center of Motion (RCM) mechanism for surgical devices, such as needle insertion. The design uses a spatial or planar mechanism to maintain the surgical tool's orientation toward the target area. The RCM operates with a single degree of freedom, based on a Parallelogram mechanism. Additionally, the modular design allows for easy replacement of components and tools, while the gear-based lock/unlock feature provides flexibility for unrestricted needle movement when necessary. This invention addresses critical gaps in current surgical device technology by offering a compact, precise, and adaptable solution tailored for modern medical applications.

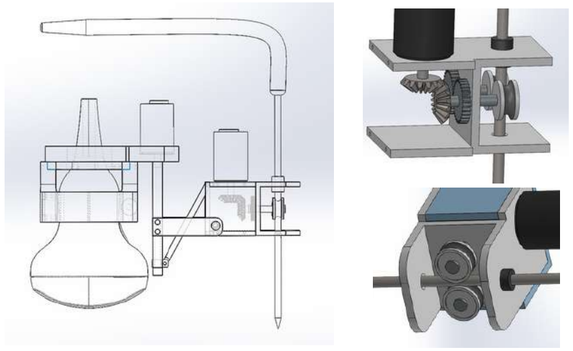


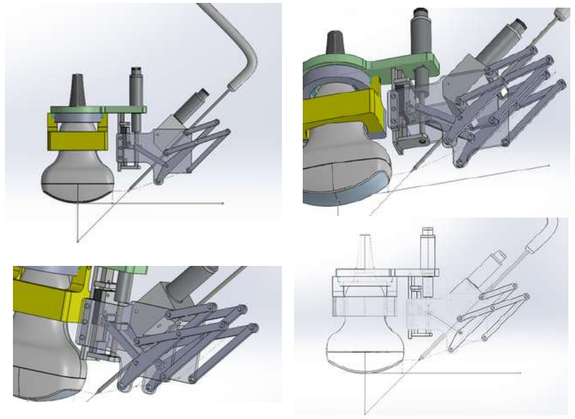
FINALISED CAD MODEL

IDEATION AND PROTOTYPES

This tool is designed for remote surgical operation and can be aligned and deployed remotely without any intervation. Needle deployment can also be disengaged for manual experimentation.

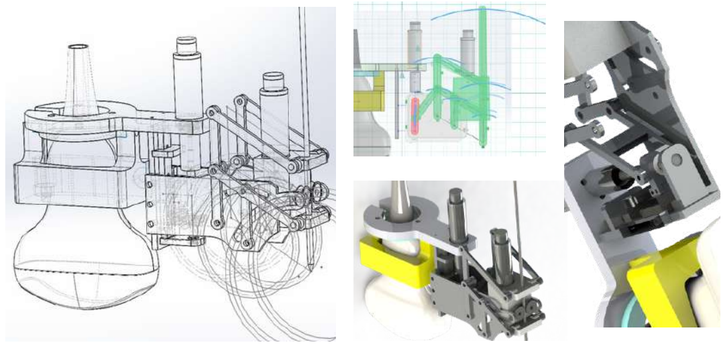
INITIAL CONCEPT





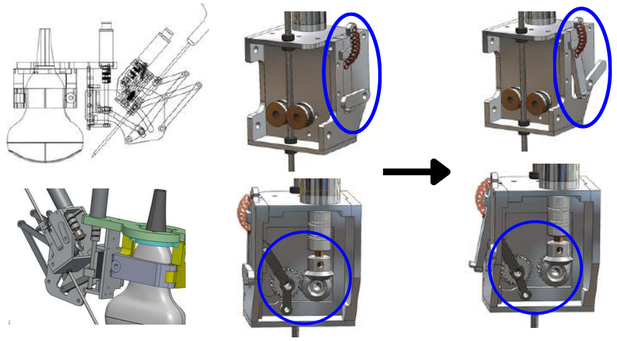
ALIGNMENT CORRECTION FOR THE DESIRED RANGE OF MOTION

INTRODUCTION OF REMOTE CENTER OF MOTION



State 1, Gear Engaged, Needle deploy ✓

State 2, Gear Disengaged, No Needle deploy, Needle freely move



WITH NEEDLE DEPLOYEMENT MECHANISM

DESIGN PARAMETERS

WORKSPACE OPTIMISATION

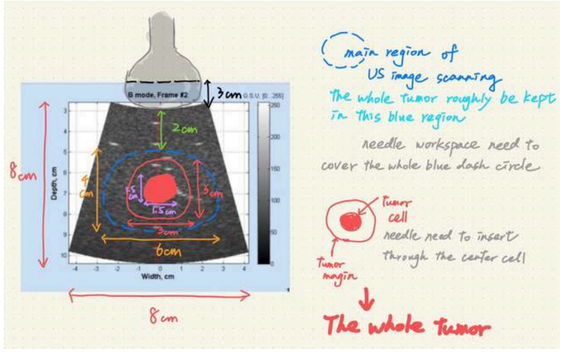


Fig. 1 REQUIRED WORKSPACE

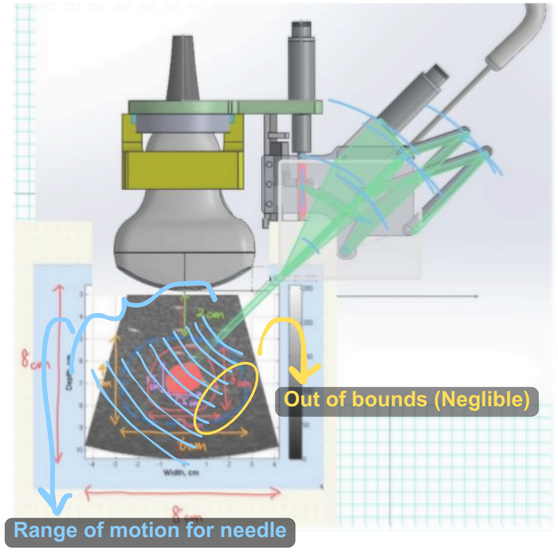


Fig. 2 ACHIEVED WORKSPACE

REMOTE CENTER OF MOTION

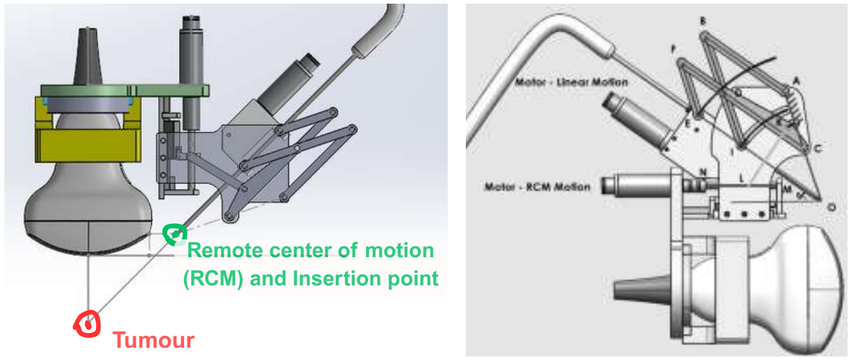


Fig. 3 Kinematic linkages of the proposed design

The design proposed uses a double parallelogram and the needle is passed through the virtual link EI, and the RCM lies on O. The link AC is fixed. The assembly to control the linear motion is moved by the attachment at E and slot to move the joint. This allows the needle to pivot around RCM point O, while the linear motion can be controlled by the motor. The parallelogram is actuated by link KL connected to link KCF.

The link KL is attached to a thread linear actuator assembly run by a motor, which moves L along the axis MN. As L moves towards M from the position in Fig 3, the assembly attains the form in Fig 4.

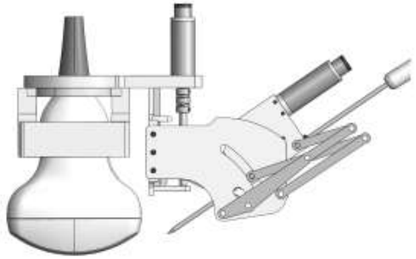
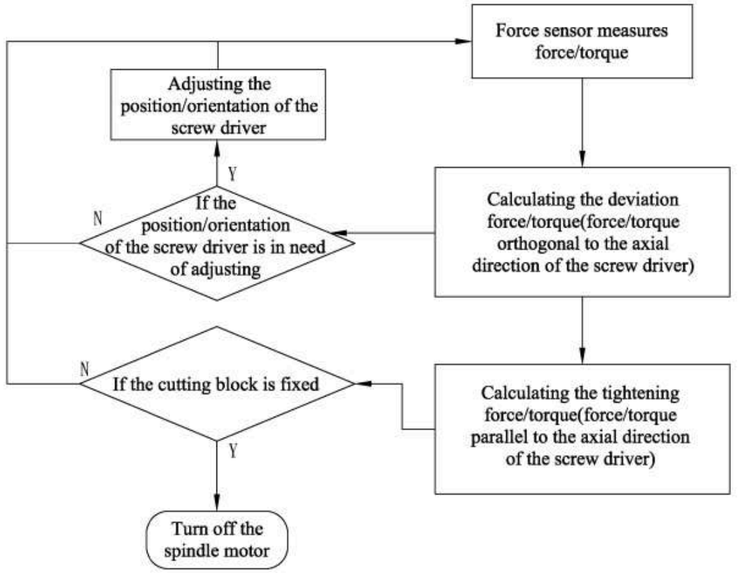
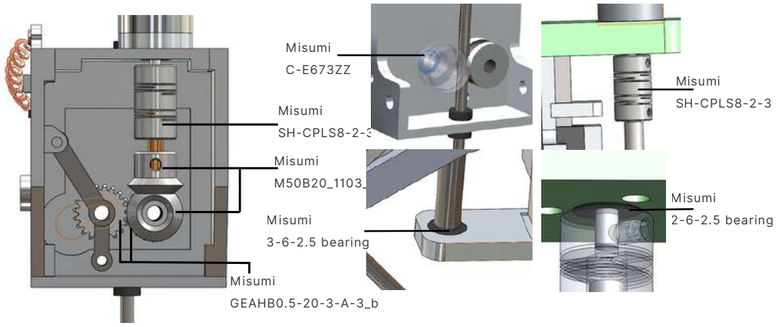


Fig. 4 Linkages at the opposite extreme position

WORKFLOW



TRANSMISSION COMPONENTS



GOVERNING EQUATION AND CONCLUSION

GOVERNING EQUATION DERIVATION

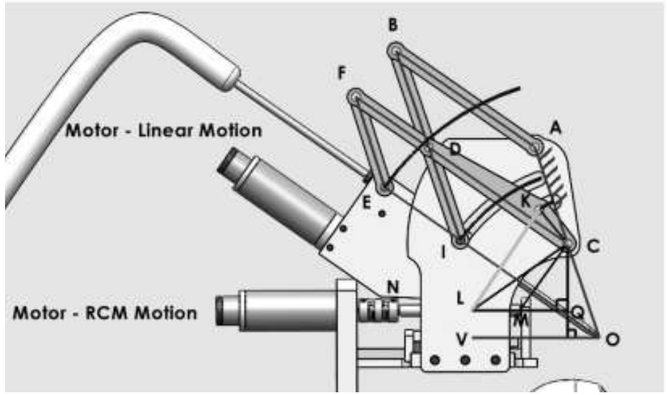


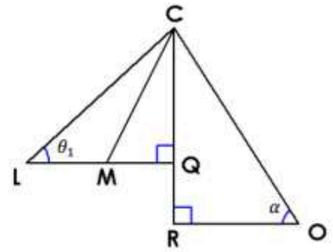
Fig 5. Linkage diagram for derivation

Defining of constant and variable parameters of the mechanism: The position of L can be varied with the rotation of the motor of the lead screw, which will change the angle of the needle (OI) from the vertical. Note that triangles in figure 6-(a), 6-(b), 6-(c), are derived from the same in figure 5.

Input length: - l_4, Output angle: - α

Mechanism Constraints (Does not change in operation)

AB	l_1
CK	l_2
KL	l_3
CQ	l_5
MQ	l_6
$\angle KCD$	β



Consider OR to be the vertical w.r.t the mechanism.

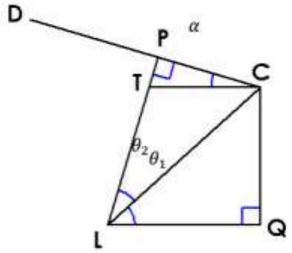
From ΔCRO and ΔCQL

$CL\cos\theta_1 = ML + MQ$

$\theta_1 = tan^{-1}\left(\frac{CQ}{ML+MQ}\right)$ ***Equation 1.***

$CL = \frac{ML+MQ}{\cos\theta_1}$ ***Equation 2.***

Fig 6(a). Derived ΔCRO and ΔCQL



From $\Delta PCT, \Delta TCL$ and ΔCQL

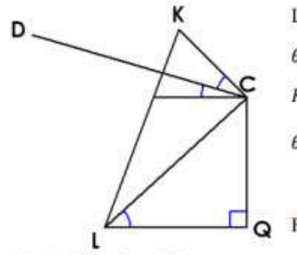
$\angle PCL = 90^0 - \theta_2$

$\angle QCL = 90^0 - \theta_1$

$(90^0 - \theta_2) - \alpha + (90^0 - \theta_1) = 90^0$

$\theta_1 + \theta_2 + \alpha = 90^0$ ***Equation 3.***

Fig 6(b). Derived $\Delta PCT, \Delta TCL$ and ΔCQL



Let $\angle KCL = \theta_3$

$\theta_3 = \beta + (90^0 - \theta_2)$ ***Equation 4.***

$KL^2 = KC^2 + CL^2 - 2.KC.KL.\cos\theta_3$

$\theta_3 = cos^{-1}\left(\frac{KC^2+CL^2-KL^2}{2.KC.KL}\right)$ ***Equation 5.***

Fig 6(c). Derived ΔKCL

Calculating α from LM

1. Find $\boldsymbol{\theta_1}$ using **LM** as the input using Eqn 1.
2. Find **CL** using Eqn 2.
3. Find $\boldsymbol{\theta_3}$ using Eqn 5.
4. Find $\boldsymbol{\theta_2}$ using Eqn 4.
5. Find $\boldsymbol{\alpha}$ using Eqn 3.

RELATION DIAGRAMS

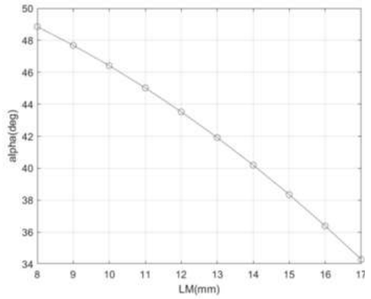
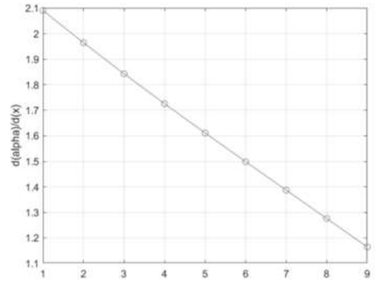


Fig 7(a). Graph: Inclination vs LM



7(b). Graph :$d\alpha/dLM$ vs No. of rot. of the motor (Pitch = 1mm)

CONCLUSION







According to the embodiment of the present invention, the system incorporates a mechanism for manually overriding automatic needle insertion. This enables the surgeon to safely and precisely push or pull the needle through the insertion point located at the Remote Center of Motion (RCM). The feature enhances modularity and safety, facilitates easy needle replacement, and provides direct force feedback to the surgeon while preserving the precision and stability offered by the RCM mechanism.